Online Appendix to

When Yield Curves Invert Together: Predicting Currency Carry Losses

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Abstract

This appendix gives the complete signal algorithm and timing conventions, derives the accounting and candidate compensation mechanism, reports supplementary estimates, and documents reproducible public-data checks, including an exploratory country-combination screen. It separates exact construction details from empirical robustness and from evidence that bears only indirectly on the mechanism.

Contents

A Signal construction and timing

This appendix gives the complete deterministic state transition used in the paper. It fixes the information set and separates fresh entry from confirmed release. It does not establish that the tenor, breadth threshold, and release length were selected independently of the return history.

A.1 Universe and information time

The nine non-dollar G10 currencies are the Australian dollar, Canadian dollar, Swiss franc, euro (Deutsche mark before 1999), British pound, Japanese yen, Norwegian krone, New Zealand dollar, and Swedish krona. The U.S. curve is excluded from the baseline count. For country i, define

$$q_{i,t} = y_{i,t}^{(10Y)} - y_{i,t}^{(2Y)}, \qquad I_{i,t} = \mathbf{1}\{q_{i,t} < 0\}.$$

All yields are end-of-month observations. A state measured at the end of month t may condition only returns realized during month $t+1$.

A.2 Curve-level transition rule

Each curve carries a live indicator $L_{i,t}$, a consecutive-steepening counter $k_{i,t}$, and an eligibility flag. A fresh inversion is a crossing from $q_{i,t-1} \geq 0$ to $q_{i,t} < 0$. The update order is entry, steepening-counter update for curves already live, release after the second consecutive increase, and finally cross-country aggregation.

Table 1: Curve-level state transition

Prior condition	Observation at t	End-of-month action
Eligible and not live	$q_{i,t-1} \geq 0$, $q_{i,t} < 0$	Set $L_{i,t} = 1$; initialize $k_{i,t} = 0$
Live	$q_{i,t} > q_{i,t-1}$	Increase $k_{i,t}$ by one
Live	$q_{i,t} \leq q_{i,t-1}$	Reset $k_{i,t} = 0$
Live; counter reaches two	Any current slope sign	Set $L_{i,t} = 0$ after the second increase
Released but still inverted	$q_{i,t} < 0$	Remain inactive and ineligible
Released and non-inverted	$q_{i,t} \geq 0$	Become eligible for a future fresh crossing
Any state	Current or lagged slope missing	Create neither a crossing nor a confirmed-steepening update

A curve can therefore be released while still inverted. It cannot re-enter until it first becomes nonnegative and then crosses below zero again. An interrupted steepening sequence resets the counter. These rules make the state depend on the path of slopes, not only on their current signs.

A.3 Cross-country aggregation and raw benchmark

After updating all curves, form

$$N_t = \sum_{i=1}^{9} L_{i,t}, \qquad S_t = \mathbf{1}\{N_t \geq 2\}.$$

The baseline state S_t is active in 92 of 458 return months and forms fifteen contiguous episodes. The corresponding current-inversion benchmark is

$$R_t = \mathbf{1}\left\{\sum_{i=1}^{9} I_{i,t} \geq 2\right\}.$$

The return in month $t+1$ is paired with S_t or, in the benchmark, R_t. The raw state asks whether fresh entry and confirmed release add information beyond the current sign of the curves.

Table 2: Incidence of the synchronized-inversion state across eras

Era	Sample months	Active months	Active share (%)
1988–94	84	33	39.3
1995–2000	72	19	26.4
2001–07	84	17	20.2
2008–09	24	6	25.0
2010–19	120	3	2.5
2020–21	24	3	12.5
2022–26	50	11	22.0

Notes: Counts reproduce the original era accounting. They describe when the state occurs, not whether its conditional-return coefficient differs across eras.

A.4 Initialization and audit outputs

At the sample start, every curve is initialized as inactive. A curve can enter only after both sides of a fresh crossing are observed. Missing values neither trigger entry nor count toward release. The recursion produces, for every country-month, the slope, raw inversion, fresh-entry flag, live status, steepening counter, release flag, and eligibility status. It then produces N_t, S_t, and episode start, end, and duration. These fields are the minimum audit trail needed to distinguish information month from return month and to reproduce the fifteen-episode count.

B Accounting and mechanism details

This appendix separates return accounting, an exposure-ordering device, and a candidate compensation friction. The first is definitional. The latter two organize the evidence but are not structurally identified by the predictive regressions.

B.1 Currency and portfolio returns

Let $s_{i,t}$ be the log dollar price of currency i and let $d_{i,t} = (i_{i,t} - i_{US,t})/12$. The paper defines the one-month return proxy and carry portfolio as

$$\begin{aligned} rx_{i,t+1} &= d_{i,t} + \Delta s_{i,t+1}, \\ rx_{t+1}^{\text{carry}} &= \frac{1}{3}\sum_{i\in\mathcal{H}_t} rx_{i,t+1} - \frac{1}{3}\sum_{i\in\mathcal{F}_t} rx_{i,t+1} \equiv C_t + X_{t+1}. \end{aligned} \tag{1}$$

The high-rate set is $\mathcal{H}_t$ and the funding set is $\mathcal{F}_t$. The construction makes $C_t > 0$ in every sample month, so the portfolio loses only when $X_{t+1} < -C_t$. Empirically, C_t averages 3.9 percent per year and has a minimum of 0.7 percent. The active-state spot-return difference is -12.85 percentage points and the total-return difference -11.31, so known income absorbs only part of the currency loss.

To organize the cross-section, write

$$\Delta s_{i,t+1} = -\lambda_i g_{t+1} + \eta_{i,t+1}, \tag{2}$$

where $g_{t+1} > 0$ is an adverse component shared across currencies, λ_i is a latent loading, and $\eta_{i,t+1}$ is idiosyncratic. A diversified long–short portfolio partly reduces idiosyncratic components; it retains the shared component when average λ_i differs between its two legs. The expanding-window equity beta $\widehat{\beta}_{i,t}$ is a predetermined proxy for λ_i, not the same primitive parameter.

B.2 Yield slopes and synchronized information

A long-minus-short yield spread combines differences in expected short-rate averages with differences in maturity-specific term premia (Campbell and Shiller, 1991; Hamilton and Kim, 2002). Consequently, one inversion is consistent with expected easing but does not reveal whether the news is domestic, global, expectations-driven, or a term-premium movement. Synchronization raises the plausibility of shared information because several markets cross a salient boundary, but it does not identify the shared shock.

The exact-one versus exact-two estimates give this idea empirical content. Exactly one live inversion is associated with carry spot and total coefficients of 8.33 and 8.78 percentage points per year. At exactly two, the coefficients are -15.72 and -15.13. The sign reversal indicates nonlinearity in cross-country breadth within the sample. It does not establish two as an optimal population threshold.

B.3 Delayed compensation as a candidate friction

Let A_{t+1} indicate an adverse event that reduces carry payoff by $J > 0$. Write

$$rx_{t+1}^{\text{carry}} = C_t - JA_{t+1} + \xi_{t+1}, \qquad \mathbb{P}_t(A_{t+1} = 1) = p_t, \qquad \mathbb{E}_t[\xi_{t+1}] = 0. \tag{3}$$

Then $\mathbb{E}_t[rx_{t+1}^{\text{carry}}] = C_t - p_t J$. For a tradable excess return in a frictionless benchmark, the Euler condition $\mathbb{E}_t[m_{t+1} rx_{t+1}^{\text{carry}}] = 0$ requires compensation when losses occur in high-marginal-utility states. Publicly visible risk therefore does not automatically imply a negative conditional mean.

The candidate friction places an information lag in compensation. Suppose

$$C_t = \bar{C}_t + \mu J \mathbb{E}_{t-1}[p_t], \qquad \mu \geq 1, \tag{4}$$

where $\bar{C}_t$ is ordinary carry income and the second component prices the current-period hazard using the earlier information set. Combining equations (3) and (4) gives

$$\mathbb{E}_t[rx_{t+1}^{\text{carry}}] = \bar{C}_t + J\left\{\mu \mathbb{E}_{t-1}[p_t] - p_t\right\}. \tag{5}$$

Hence

$$\mathbb{E}_t[rx_{t+1}^{\text{carry}}] < 0 \quad \iff \quad p_t > \mu \mathbb{E}_{t-1}[p_t] + \frac{\bar{C}_t}{J}. \tag{6}$$

The inequality is the model's economic content. A rise in perceived risk is insufficient: the current expected loss must exceed both previously embedded event compensation and ordinary carry income. Fresh synchronized inversions may mark such a belief innovation, but the paper does not observe p_t or identify the adjustment friction.

B.4 Conditional uncovered interest parity

The supporting Fama regression estimates the appreciation slope separately by state:

$$\Delta s_{i,t+1} = a_i^{(s)} + \phi_s d_{i,t} + u_{i,t+1}, \qquad S_t = s, \quad s \in \{0, 1\}. \tag{7}$$

Under the paper's quote convention, uncovered interest parity implies $\phi_s = -1$. The pooled estimate is 0.50 outside the state and -1.33 inside it. The point estimates say that high-rate currencies depreciate in the state in which carry loses. The state interaction has a Newey–West t-statistic of -2.39 but an episode-bootstrap p-value of 0.340. The pooled panel contains many currency-months, whereas the common state changes across only fifteen episodes; the episode result is therefore the appropriate warning against treating -1.33 as a precise structural slope.

B.5 Exploratory curve-shape compensation

Define the cross-country average yield at maturity m as $\bar{y}_t^{(m)} = 9^{-1} \sum_{i=1}^{9} y_{i,t}^{(m)}$. An active-state month is classified as downcurve when

$$\bar{y}_t^{(3M)} > \bar{y}_t^{(2Y)} > \bar{y}_t^{(10Y)}. \tag{8}$$

There are 52 downcurve months and 35 overlap the synchronized state. In those 35 months, carry income averages 6.48 percent per year and the spot leg -0.96; in the other 57 active-state months, income averages 4.31 and the spot leg −15.55. Total carry returns average 5.51 percent in downcurve-state months and −11.24 percent in the remaining state months.

This split is economically suggestive: high, fully downward-sloping curves can mark inflation normalization with unusually large carry income, whereas other synchronized inversions can look more like uncompensated stress. The classification was developed after examining apparent false alarms and is concentrated in 1988–91 and 2023–24, plus one month in 2007. Rate level, inflation, and curve shape move together in those periods, so the heterogeneity is exploratory and does not establish a feasible trading rule.

C Supplementary evidence and robustness

All exercises in this appendix use the same historical sample on which the state was assessed. They measure sensitivity and economic breadth; they do not create a frozen out-of-sample test.

C.1 Distribution and episode influence

Removing each portfolio's full-sample bottom-decile months leaves negative carry spot and total state effects. Because the deletion conditions on the realized outcome, it diagnoses tail concentration rather than estimating a preferred conditional mean.

Table 3: State effects after removing bottom-decile months

Portfolio component	Remaining signal months	Difference (%/yr)	Episode wild p
Carry spot	77	-7.24	0.006
Carry total	77	-5.50	0.024
High-beta spot	77	-3.72	0.084
High-beta total	78	-3.68	0.060

Notes: Each regression drops that return series' full-sample bottom-decile months from both states. The exercise asks whether a few crash observations mechanically generate the conditional-mean result; it does not address selection among alternative signal definitions. Wild-bootstrap p-values cluster by IYC episode.

The median and every reported lower quantile also move left. For carry spot returns, the monthly median changes from 0.35 percent outside the state to −0.76 percent inside, the 25th percentile from −0.93 to −1.98, and the 10th percentile from −2.35 to −3.94. This evidence is more informative about a broad distributional shift than the fact that all five worst months occur under the lagged state.

Leave-one-episode-out carry estimates remain within roughly one-third of the full-sample coefficient. The high-beta effect remains statistically detectable under every one-episode deletion, with a largest reported bootstrap p-value of 0.006. These checks reduce dependence on a single episode but do not overcome the limited number of episodes.

C.2 Calendar stability

The carry-spot coefficient is negative in both calendar halves. The high-beta result is less stable and is concentrated after 2005. The split is descriptive because the rule was not frozen on the first half before evaluation on the second.

Table 4: The headline estimates in two halves of the sample

	1988:01–2004:12			2005:01–2026:02		
Series	b	$t_{episode}$	wild p	b	$t_{episode}$	wild p
Carry spot	-11.41	-2.54	0.037	-15.85	-1.99	0.010
Carry total	-10.20	-2.04	0.052	-14.76	-1.84	0.047
High-beta spot	-4.01	-1.12	0.287	-11.91	-4.07	0.010
High-beta total	-3.03	-0.80	0.448	-11.54	-3.85	0.011
Months		204			254	
Signal months		54			38	
Episodes		7			8	

Notes: Each column reports the state regression $y_t = a + b\,\mathbf{1}\{\text{IYC}\}_{t-1} + e_t$ estimated separately by calendar half. Coefficients are annualized percentage points; $t_{episode}$ clusters by IYC episode, and wild-bootstrap p-values use $B = 9{,}999$. This is a stability check, not a frozen out-of-sample test.

C.3 State-definition sensitivity

The original grids vary breadth, release confirmation, and the short maturity. Exactly one live inversion has positive carry coefficients, while exactly two produces the largest negative coefficients. At-least-three and at-least-four rules have fewer active months and later coverage. Releasing after one steepening month is noisier; requiring three retains the state longer. Rebuilding the state with a 10Y–3M rather than 10Y–2Y slope preserves the broad adverse ordering but changes episode timing.

The raw current-inversion state is active in 165 months, compared with 92 for the live state. It captures fewer of the worst carry months and has smaller return coefficients. The live state can retain losses that arrive after curves begin to re-steepen, but its additional tail coverage is partly mechanical because the historical crash chronology helped motivate persistence.

A reciprocal-age version of the current-inversion count separates two timing margins. It has a high-beta spot coefficient of -14.87 ($p = 0.001$), larger than the baseline live state's -7.53, but a smaller carry coefficient of -10.43 ($p = 0.040$) than the baseline's -12.85. Under the full control set, the age-weighted high-beta coefficient remains -13.82 ($p = 0.008$), whereas its carry coefficient attenuates to -7.55 ($p = 0.084$). These in-sample comparisons suggest that exposure repricing is concentrated near fresh information, while the path-dependent release rule is more important for carry losses that continue into re-steepening.

Table 5: Measurement perturbations and exposure ordering

State construction	Carry spot	Wild p	High-beta spot	Wild p
10Y–2Y baseline	−12.85	0.001	−7.53	0.010
10Y–3M slope	−4.68	0.146	−5.25	0.011
U.S.-term-premium-adjusted 10Y–2Y	−5.42	0.107	−6.17	0.004

Notes: Coefficients are annualized percentage-point active-state differences. The 10Y–3M row rebuilds every curve state using a different short maturity. The term-premium row removes each national slope's fitted exposure to the U.S. ACM 10Y-minus-2Y slope premium before rebuilding the state. Both perturbations attenuate carry while retaining the beta-sorted exposure result.

Table 6: State-definition multiverse: reported and untested dimensions

Dimension	Reported neighborhood	Remaining uncertainty
Breadth	At least 1–4; exactly 1–2	Frequency-matched and country-combination rules
Release	One-, two-, and three-month confirmation	Other persistence and duration-matched placebo rules
Tenor	10Y–2Y baseline; 10Y–3M check	Other maturities and inversion cutoffs
Freshness	Fresh crossing versus current inversion	Alternative age decay and entry delay
Geography	Non-U.S. G10 baseline; U.S.-inclusion checks	Other developed-market sets and country combinations
Portfolio	Three-by-three carry and beta sorts	Other leg sizes, volatility scaling, feasible forwards
Timing	Month-end recursion and one extra lag	Vendor timestamps, asynchronous closes, vintage data

C.4 Exposure breadth and composition

The carry and beta portfolios hold persistent characteristics rather than changing membership only when the state turns on. New Zealand occupies the high-rate leg in 85 percent of all months and 91 percent of state months; Australia in 71 and 73 percent. Japan and Switzerland remain the principal funding currencies. This makes the state result more naturally a repricing of standing exposure than a mechanical change in portfolio composition.

The seven floating emerging-market currencies are not used to construct the G10 state. All seven bilateral state coefficients are negative; pooled spot and total effects are −14.37 and −13.35 percentage points per year, with episode-bootstrap p-values of 0.104 and 0.079. This is directional breadth, not an independent replication, because the same global dates define the state and the emerging-market sample is shorter.

The common-sample comparison separates international spot repricing from compensation.

Table 7: Spot repricing and compensation across currency universes, 1996–2026

Universe	Return leg	Inactive mean	Active mean	State difference	Episode wild p
G10	Spot	2.96	−12.79	−15.75	< 0.001
G10	Total	6.22	−8.67	−14.89	0.003
G10+EM	Spot	0.46	−12.51	−12.97	0.176
G10+EM	Total	11.68	2.51	−9.18	0.235

Notes: Annualized percentages over the common 1996:01–2026:02 sample. The G10+EM universe adds seven floating emerging-market currencies and applies the same three-by-three rate sort. Active-state income implied by total minus spot is approximately 4.12 percent for G10 and 15.02 percent for G10+EM. The nearly equal spot means but different total means show how predetermined carry income changes the payoff from similar currency repricing.

All reported exposure tests use expanding S&P 500 betas. An oil-beta sort adds little independent variation because its return is largely spanned by the carry and equity-beta portfolios.

C.5 Conditional UIP and macro timing

Appendix B gives the conditional Fama specification. The pooled slope is 0.50 outside the state and −1.33 inside, but the state interaction has episode-bootstrap $p = 0.340$. The point estimate is consistent with high-rate currencies depreciating during the state, while the interaction remains imprecise at the episode level.

The leading indicator declines by about 0.6 percent over six months, with similar magnitudes around episode onsets and under lag controls. Industrial production declines about 1.8 percent at six months in the baseline projection but becomes imprecise with the fullest controls. The leading-indicator response is persistent across these specifications; the industrial-production response is not.

C.6 Benchmarks, controls, and term premia

The U.S. inversion, average G10 slope, and its change do not span the synchronized state. Lagged NFCI and VIX controls also leave the carry coefficient near its baseline magnitude. Contemporaneous dollar and FX-volatility controls show that the realized loss is not simply the average dollar move or the measured volatility innovation, but they are not predictive controls.

The term-premium adjustment is asymmetric. After removing fitted exposure to the U.S. ACM 10Y-minus-2Y term-premium slope, 83 percent of national inversions remain. The high-beta coefficient is −6.2 percentage points with $p = 0.004$. The carry coefficient is about −5.4 with $p = 0.11$, roughly half the baseline magnitude. A common U.S. term-premium component therefore does not eliminate exposure ordering, but may explain part of the carry association. Comparable foreign-country term-premium decompositions are unavailable.

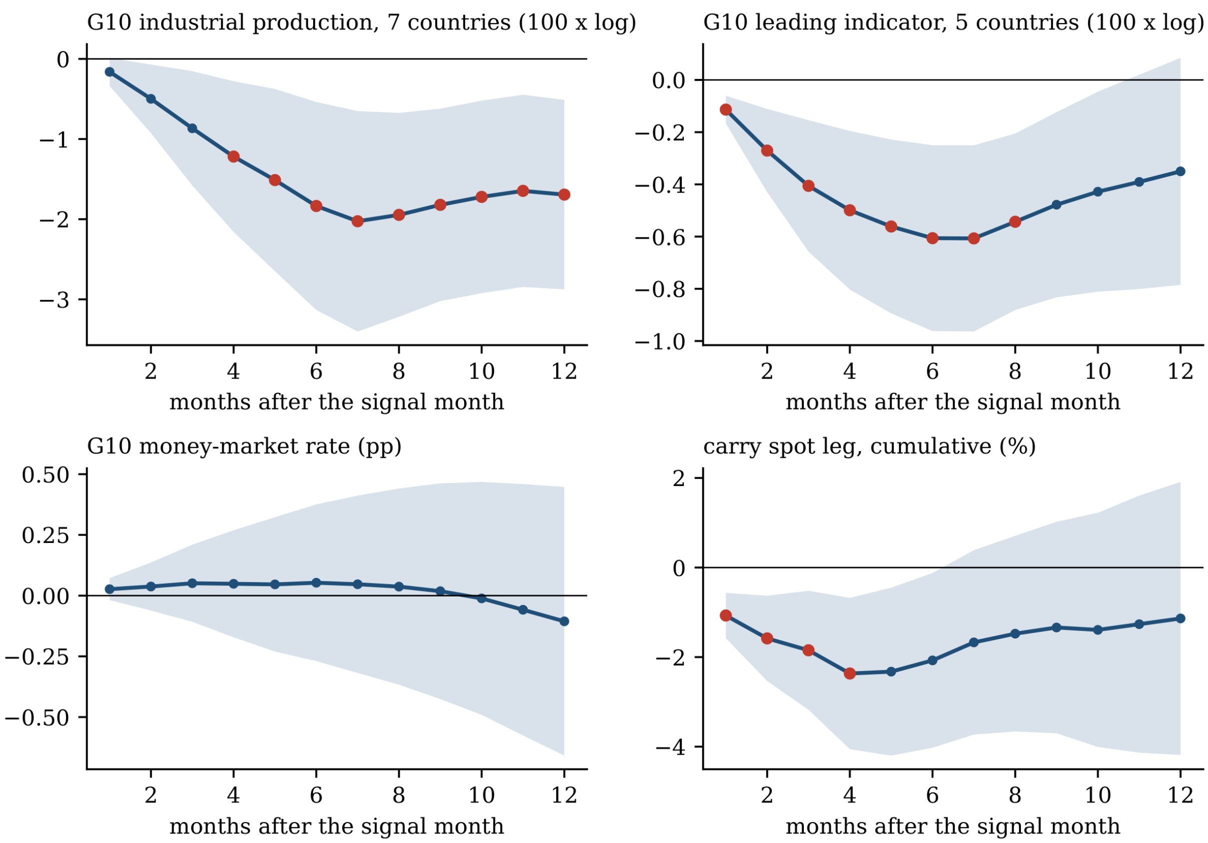

Figure 1: Outcomes following a synchronized-inversion state. The panels show local projections of cumulative changes in G10 industrial production, the OECD composite leading indicator, the average G10 money-market rate, and the carry spot leg. Bands are 90-percent Newey–West intervals; markers denote episode-bootstrap $p < 0.10$. Overlapping horizons and persistent states make these supporting temporal associations rather than causal responses.

C.7 Interpretive boundary

The conditional return gap, distributional shift, and exposure ordering are statistically detectable in the reported specifications. The conditional-UIP interaction and industrial-production response are imprecise; emerging-market breadth and the downcurve split are exploratory. Feasible forward returns, complete data-vintage auditing, a comprehensive state multiverse, and a frozen future sample remain untested.

D Public-data mechanism checks

This appendix uses public data to examine outcomes around synchronized delivered policy easing. The event is intentionally later than the yield-curve state: inversions can encode expected easing, whereas policy-rate cuts record policymakers' actions. The exercise tests selected mechanism implications but, because it observes policy delivery rather than the predictive state, it does not validate the inversion classifier.

D.1 Event construction

The sample is January 1988 through December 2025. Let $r_{i,t}^{pol}$ be country i's BIS end-of-period policy rate and define

$$K_{i,t} = \mathbf{1}\{\Delta r_{i,t}^{pol} \leq -0.10\}, \qquad B_t = \sum_i K_{i,t}, \tag{9}$$

where rates and the 10-basis-point cutoff are in percentage points. Let $D_t = 1$ when $B_t \geq 3$ and at least six country rates are observed. The onset indicator is

$$O_t = D_t \prod_{\ell=1}^{3} (1 - D_{t-\ell}). \tag{10}$$

Thus an onset requires at least three cutting countries after three months without another synchronized-cut month. The rule produces fifteen onsets. These fifteen public easing onsets are not the paper's fifteen inversion episodes; the identical counts are coincidental.

The public currency outcome is a spot-only proxy. BIS exchange rates are signed so that a positive return is foreign-currency appreciation against the dollar. At $t-1$, currencies are ranked by their policy-rate differentials against the United States; the proxy is long the three highest-ranked and short the three lowest-ranked. The rank is predetermined, but O_t is observed during month t, so month-t returns are contemporaneous event associations rather than feasible predictions made before easing.

Additional outcomes are CFTC non-commercial net positions scaled by open interest, the New York Fed ACM 10-year expected-rate and term-premium components (Adrian et al., 2013), the OECD G7 composite leading indicator, NFCI, and VIX. CFTC coverage is limited to seven directly matchable contracts and uses a documented Deutsche-mark/euro convention. ACM is a U.S.

diagnostic, not a decomposition of the nine foreign slopes. The macro and financial-condition series are current vintage.

D.2 Event estimands and inference

The code uses two estimands. For a level outcome Y, the horizon-h event response is

$$\widehat{\theta}_h^{\text{change}} = \frac{1}{n_h} \sum_{j \in \mathcal{J}_h} \left(Y_{\tau_j + h} - Y_{\tau_j - 1} \right). \tag{11}$$

For a monthly return flow R, it is

$$\widehat{\theta}_h^{\text{return}} = \frac{1}{n_h} \sum_{j \in \mathcal{J}_h} \sum_{\ell=0}^{h} R_{\tau_j + \ell}, \tag{12}$$

where τ_j indexes valid onset dates. Ninety-percent intervals resample onset episodes; leave-one-onset ranges diagnose influence. The reference distribution enumerates circular shifts of the complete onset sequence, retains shifts with the observed number of valid event outcomes, includes the observed assignment, and doubles the smaller inclusive-tail rank. With irregular missingness, the retained shifts need not form a transformation group; the resulting values are conditional finite-rotation references, not exact causal randomization tests.

Six declared primary outcomes receive Holm adjustment: the shadow-carry spot return over months 0–1; the CFTC positioning change through month 3; ACM expected-rate and term-premium changes through month 3; the CLI change through month 6; and the NFCI change through month 3. The declaration is recorded in the analysis configuration and run manifest, but not in an immutable pre-analysis registration. VIX and other horizons are descriptive.

The signs mostly fit a broad stress-and-easing chronology: the shadow carry proxy depreciates, carry-aligned positions decline, the expected-rate component falls, and the leading indicator weakens. Estimates are imprecise, and no primary outcome survives the Holm adjustment. The U.S. term-premium estimate is nearly flat at three months. These results are directional mechanism evidence around policy delivery, not evidence that the inversion state predicts them.

Table 8: Outcomes around synchronized delivered-easing onsets

Outcome	Estimate	90% CI	RI p	Holm p	Rotations
Policy-ranked shadow carry spot return, months 0–1	-1.10	[-3.55, 1.18]	0.195	0.975	441
Carry-aligned CFTC non-commercial net share, change through month 3	-4.39	[-13.53, 5.40]	0.741	1.000	81
ACM 10-year expected-rate component, change through month 3	-0.13	[-0.31, 0.07]	0.202	0.975	396
ACM 10-year term premium, change through month 3	-0.03	[-0.19, 0.14]	0.980	1.000	396
OECD G7 CLI, change through month 6	-0.34	[-0.72, 0.00]	0.159	0.952	353
Chicago Fed NFCI, change through month 3	0.03	[-0.08, 0.15]	0.520	1.000	396
VIX, change through month 1	1.89	[-2.30, 5.81]	0.400	–	20

Notes: Events are synchronized delivered-easing onsets constructed from public BIS policy rates. They are a downstream policy-rate state, distinct from the paper's predictive IYC state. RI p-values enumerate every circular rotation, condition on the observed number of valid episode outcomes, include the observed assignment, and double the smaller inclusive tail rank. Rotations reports the resulting finite reference count. The CFTC reference is small because missing contract coverage makes most rotations change the valid event count; its RI result is correspondingly coarse. With irregular missingness, the same-count rotations need not form a transformation group, so these are conditional finite-rotation references rather than exact causal randomization tests. Intervals resample onset episodes. Holm p-values adjust the six prespecified primary outcomes. Currency returns and CFTC positioning are percentage points, with CFTC positions measured as a share of open interest; ACM components are percentage points; CLI, NFCI, and VIX are index points.

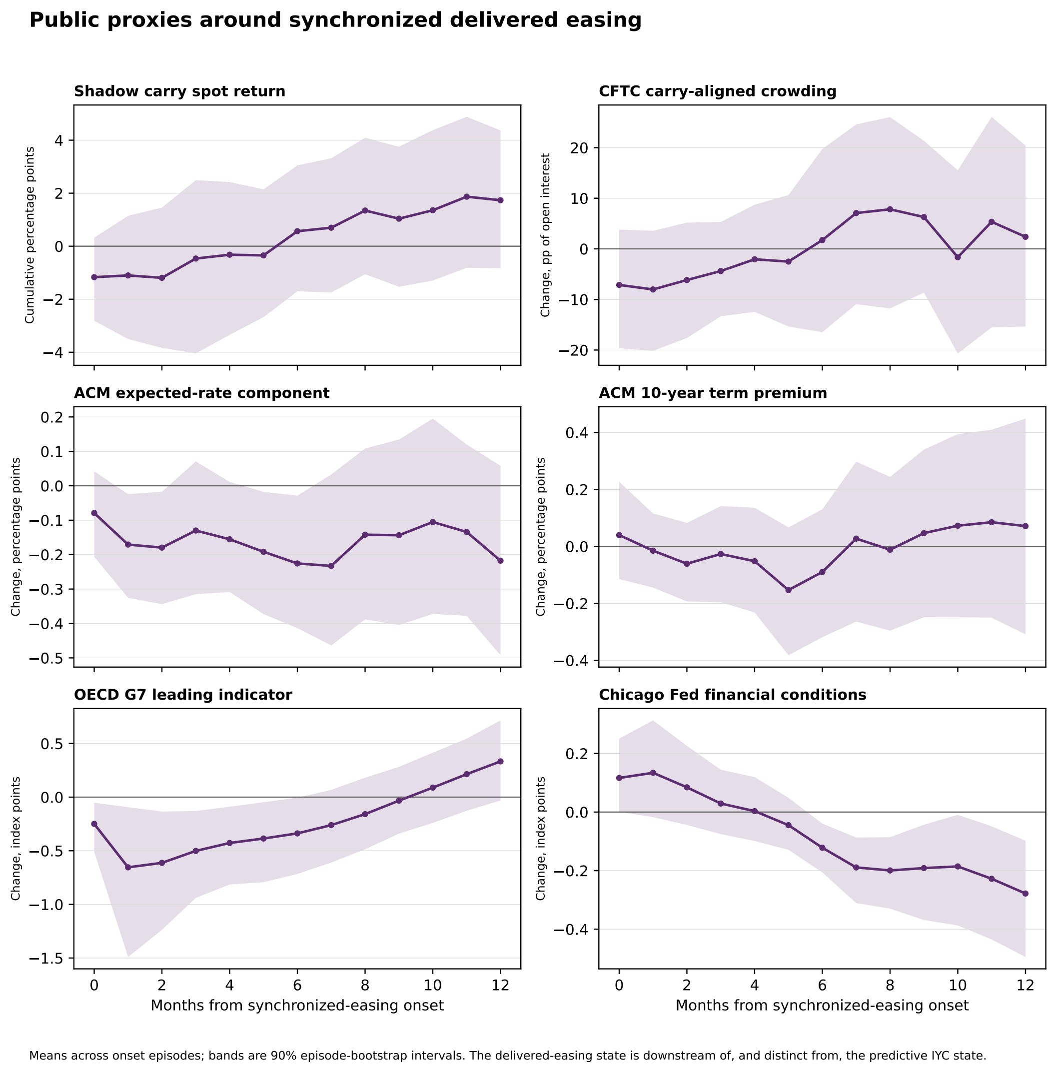

Figure 2: Outcomes around synchronized delivered-easing onsets. Event time zero is a month in which at least three policy rates fall by at least 10 basis points after a three-month quiet period. Panels show the code-defined cumulative return or endpoint-change estimands, with 90-percent episode-bootstrap intervals.

D.3 Sensitivity and interpretation

The shadow-carry response is recomputed using two- and four-country event thresholds and after omitting each event currency. Every threshold recomputes its own quiet period, so onset counts need not be monotone in the required number of cutting countries.

The exercise has four binding limits. Delivered easing is endogenous to prior information and occurs downstream of the paper's predictor. Policy-rate ranks are not money-market or forward rates, and spot returns omit carry income, basis, transaction costs, and implementation lags. CFTC, ACM, and current-vintage macro series are imperfect proxies for the intended objects. Finally, fifteen onsets—fewer for some outcomes—leave low episode-level power. The public results are therefore useful as a reproducible challenge to mechanism implications, not as a replication of

Table 9: Public-proxy sensitivity: shadow carry spot return

Construction	Estimate	RI p	Events	Rotations
At least three countries easing	-1.10	0.195	15	441
At least two countries easing	-1.24	0.060	22	434
At least four countries easing	-0.46	0.635	18	438
Exclude AUD from event rule	-0.50	0.556	17	439
Exclude CAD from event rule	0.36	0.562	18	438
Exclude CHF from event rule	-1.21	0.150	15	441
Exclude EUR from event rule	-1.00	0.214	16	440
Exclude GBP from event rule	-0.81	0.321	20	436
Exclude JPY from event rule	-0.27	0.787	19	437
Exclude NOK from event rule	-1.65	0.063	13	443
Exclude NZD from event rule	-0.53	0.580	15	441
Exclude SEK from event rule	-0.47	0.606	17	439

Notes: Each threshold definition recomputes its own three-month quiet window. Onset counts therefore need not be monotone in the number of countries required to cut. Leave-one-currency rows omit that currency from the event rule.

the headline state.

E Which countries matter in the public proxy?

This appendix asks whether particular country pairs or triples dominate the public delivered-easing association. It does not observe the original nine-country yield-slope panel and therefore cannot identify which curves created the paper's fifteen inversion episodes. The analysis uses the same BIS policy-rate and exchange-rate data as Appendix D.

E.1 Combination screen

All 36 currency pairs and 84 triples are enumerated. For a given combination, an onset occurs when every member cuts its policy rate by at least 10 basis points after three months without that same joint cut. The outcome is the public shadow-carry spot return accumulated over event months zero and one. The ranking of all nine currencies uses lagged policy-rate differentials; the tested country combination defines the event, not the portfolio.

Inference is reported only for combinations with at least six valid events. Every eligible event sequence is shifted through every circular calendar rotation. Raw two-sided rotation p-values are supplemented with a common-rotation maximum absolute standardized statistic that controls family-wise error across every eligible pair and triple. Episode-resampling intervals, leave-one-event means, and alternative minimum-event rules diagnose small-sample fragility.

Table 10: Country combinations in the public delivered-easing proxy

Combination	Events	Mean $[0,1]$	Raw p	Family-adjusted p
CHF–GBP	6	−4.63	0.009	0.042
AUD–EUR	6	−4.06	0.009	0.088
GBP–SEK	9	−3.28	0.018	0.152
CHF–SEK	7	−2.74	0.070	0.549
CHF–NOK	8	−2.62	0.057	0.538
CAD–NOK–NZD	7	−4.29	0.013	0.055
NOK–NZD–SEK	6	−3.62	0.026	0.279
CAD–GBP–NOK	6	−3.58	0.022	0.277
GBP–NZD–SEK	8	−3.52	0.018	0.143
AUD–NOK–NZD	6	−3.11	0.044	0.440

Notes: The table reports the five most negative eligible pairs and five most negative eligible triples. An event requires every listed country to cut its BIS policy rate by at least 10 basis points after three months without the same joint cut. The outcome is the cumulative policy-ranked shadow-carry spot return in event months 0 and 1, in percentage points. Combinations require at least six valid events. Raw two-sided p-values use every circular timing rotation. The final column controls family-wise error across all 42 eligible pairs and triples with a common-rotation maximum absolute standardized statistic. This is a delivered-easing proxy, not a replication of the yield-curve state.

Of 120 combinations, 42 have at least six events. CHF–GBP is the only 5-percent family-wide result: six joint-cut onsets have a two-month mean shadow-carry spot return of -4.63 percentage points, raw $p = 0.009$, and adjusted $p = 0.042$. The 90-percent episode-bootstrap interval is $[-9.52, 0.26]$, however, and deleting October 2008 reduces the mean to -2.62. CAD–NOK–NZD is next, with seven events, a -4.29-point mean, and adjusted $p = 0.055$.

The family result depends on admitting sparse combinations. With a six-event minimum, 42 combinations are eligible and one clears 5 percent. With minimums of eight or ten events, 25 and 15 combinations remain, no adjusted result clears 5 percent, and the smallest adjusted p-values are 0.105 and 0.224. The evidence does not establish a stable country bloc.

E.2 Sensor countries versus loss-bearing currencies

The public lead does not match the baseline loss-bearing cross-section. CHF and GBP have near-zero or modest equity betas and small, imprecise state-conditioned currency losses. AUD and NZD bear the larger losses. The own-curve horse race also shows that a country's own curve generally does not forecast its currency after conditioning on the synchronized state.

This contrast suggests a testable distinction: countries whose bond or policy states reveal a shared episode need not be the currencies that bear its losses. Joint Swiss–U.K. easing may function as a calendar sensor for broad stress, while high-rate, high-beta currencies absorb the carry loss. The distinction is economically consistent with synchronization and exposure ordering, but the present result is only hypothesis-generating because delivered easing is downstream of forward-looking yield-curve inversions.

The direct test requires the nine-country live-state matrix. A fifteen-episode-by-nine-country contributor ledger could enumerate which curves activate each episode, then compare contributor combinations with the subsequent currency-loss cross-section under family-wide inference. The public combination screen specifies that future test but does not decompose the headline result.

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